

Explicit Formulas for Cyclotomic Multiple Bernoulli Numbers

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- ① Background and motivation
- ② Main result
- ③ Application to p -adic special values
- ④ Appendix

Background and motivation

p -adic multiple L -functions

Furusho, Komori, Matsumoto, and Tsumura introduced the p -adic multiple L -functions

$$L_{p,r}((s_i)_r; (\omega^{k_i})_r; (1)_r; c) := \int_{(\mathbb{Z}_p^r)'} \langle x_1 \rangle^{-s_1} \langle x_1 + x_2 \rangle^{-s_2} \cdots \langle x_1 + \cdots + x_r \rangle^{-s_r} \omega^{k_1}(x_1) \omega^{k_2}(x_1 + x_2) \cdots \omega^{k_r}(x_1 + \cdots + x_r) \prod_{i=1}^r d\tilde{m}_c(x_i),$$

which are multivariable analogues of the Kubota–Leopoldt p -adic L -functions $L_p(s; \omega^k)$. For the precise definition, see the Appendix.

Work of Furusho and Jarossay

Furusho and Jarossay studied the special values of the p -adic multiple L -functions and proved the following theorem.

Theorem (Furusho–Jarossay, 2020)

For any r -tuple $(n_i)_r$ of positive integers, the family

$$\left(p^{\sum_{i=1}^r n_i} L_{p,r}((n_i)_r; (\omega^{-n_i})_r; (1)_r; c) \right)_{p \in \mathcal{P}_c}$$

is expressed as

$$\sum_{l=(l_i)_r \in \mathbb{N}_0^r} \sum_{\substack{\epsilon \in \mu_c \\ (\epsilon_i)_r \in (\mu_c \setminus \{1\})^r}} \sum_{J=(P_1, P_2, P_3) \in E_r} \sum_{\substack{\xi \in \mu_c \\ \prod_{j \in P_3} \left(\frac{\epsilon_j^{(P_2, 3)}}{\epsilon_j} \right)^{-\kappa'_j} = \epsilon}} \frac{\epsilon}{\prod_{i=1}^r (1 - \epsilon_i)} \prod_{i=1}^r \binom{-n_i}{l_i} \mathcal{B}_{0, \xi}^{(l_J, \epsilon_J, \kappa_J)} \sum_{\delta \in \Delta(T_{r, J})} \mathfrak{H}(w(l)_\delta)^{\text{Frob}^{-1}}.$$

Work of Furusho and Jarossay

In the theorem on the previous slide, we use the following abbreviations:

- p MLFVs denotes the special values of the p -adic multiple L -functions $L_{p,r}((n_i)_r; (\omega^{-n_i})_r; (1)_r; c)$.
- CMBNs denotes the cyclotomic multiple Bernoulli numbers $\mathcal{B}_{0,\xi}^{(l_J, \epsilon_J, \kappa_J)}$.
- CMHVs denotes the cyclotomic multiple harmonic values $\mathfrak{H}(w(l)_\delta)$.

Thus, the theorem of Furusho and Jarossay can be written as

$$p\text{MLFVs} = \sum (\text{CMBNs}) \times (\text{CMHVs}).$$

The CMBNs are introduced by Furusho and Jarossay in their paper and are the main objects of this talk.

Cyclotomic multiple harmonic sums (CMHSs)

To define CMBNs, we recall the definition of CMHSs first.

Definition

Let

$$\begin{aligned} c &\in \mathbb{Z}_p^\times \cap \mathbb{N}_{\geq 2}, & (l_i)_r &= (l_1, \dots, l_r) \in \mathbb{N}_0^r, \\ (\epsilon_i)_r &= (\epsilon_1, \dots, \epsilon_r) \in \mu_c^r, & h &\in \mathbb{N}, \\ (\kappa_i)_{r-1} &= (\kappa_1, \dots, \kappa_{r-1}) \in \mathbb{N}_0^{r-1}. \end{aligned}$$

We define the *variant of cyclotomic multiple harmonic sums* as

$$\begin{aligned} \mathcal{S}_{(\kappa_i)_{r-1}, h}((l_i)_r; (\epsilon_i)_r) = \\ \sum_{\substack{(u_1, \dots, u_r) \in \mathbb{N}_0^r, \ u_1 < h \\ \forall i \geq 2, \ u_{i-1} + \kappa_{i-1} h < u_i \\ \forall i \geq 2, \ u_i < u_{i-1} + (\kappa_{i-1} + 1)h}} \left(\frac{\epsilon_2}{\epsilon_1}\right)^{u_1} \cdots \left(\frac{1}{\epsilon_r}\right)^{u_r} u_1^{l_1} \cdots u_r^{l_r} \in \mathbb{Q}(\mu_c), \end{aligned}$$

where we formally let $0^0 = 1$.

Cyclotomic multiple Bernoulli numbers (CMBNs)

Definition

Let

$$\begin{aligned} c &\in \mathbb{Z}_p^\times \cap \mathbb{N}_{\geq 2}, & (l_i)_r &= (l_1, \dots, l_r) \in \mathbb{N}_0^r, \\ (\epsilon_i)_r &= (\epsilon_1, \dots, \epsilon_r) \in \mu_c^r, & h &\in \mathbb{N}, \\ (\kappa_i)_{r-1} &= (\kappa_1, \dots, \kappa_{r-1}) \in \mathbb{N}_0^{r-1}, & \xi &\in \mu_c, \\ l &\in \{0, \dots, l_1 + \dots + l_r + r\}. \end{aligned}$$

We define the *variant of cyclotomic multiple Bernoulli numbers*

$$\mathcal{B}_{l,\xi}^{((l_i)_r; (\epsilon_i)_r; (\kappa_i)_{r-1})} \in \mathbb{Q}(\mu_c)$$

by

$$\mathcal{S}_{(\kappa_i)_{r-1}, h}((l_i)_r; (\epsilon_i)_r) = \sum_{\substack{0 \leq l \leq l_1 + \dots + l_r + r \\ \xi \in \mu_c}} \mathcal{B}_{l,\xi}^{((l_i)_r; (\epsilon_i)_r; (\kappa_i)_{r-1})} h^l \xi^h.$$

Recursive formula for CMBNs

Furusho and Jarossay provided the following recursive relation.

$$\begin{aligned} \mathcal{S}_{(\kappa_i)_r, h}((l_i)_{r+1}, (\epsilon_i)_{r+1}) = & \sum_{\substack{0 \leq l \leq l_{r+1}+1 \\ \xi \in \mu_c \\ 0 \leq \tilde{l} \leq l}} \mathcal{B}_{l, \xi}^{(l_{r+1}+1; \epsilon_{r+1}; \emptyset)} \binom{l}{\tilde{l}} h^{l-\tilde{l}} \left\{ (\kappa_r + 1)^{l-\tilde{l}} \xi^{(\kappa_r+1)h} - \kappa_r^{l-\tilde{l}} \xi^{\kappa_r h} \right\} \\ & \times \mathcal{S}_{(\kappa_i)_{r-1}, h}((l_i)_{r-1}, l_r + \tilde{l}; (\xi^{-1} \epsilon_r^{-1} \epsilon_i)_r) \\ & - \sum_{\tilde{l}=0}^{l_{r+1}} \binom{l_{r+1}}{\tilde{l}} (\kappa_r h)^{l_{r+1}-\tilde{l}} \epsilon_{r+1}^{-\kappa_r h} \mathcal{S}_{(\kappa_i)_{r-1}, h}((l_i)_{r-1}, l_r + \tilde{l}; (\epsilon_i)_r). \end{aligned}$$

By substituting the expressions for the CMHSs into this identity and comparing coefficients, one obtains a recurrence relation for the CMBNs. Thus, the CMBNs can be computed recursively, but the computation becomes increasingly complicated as the depth r increases. Therefore, the aim of this work is to find an explicit formula for the CMBNs in terms of the known quantities.

Main result

Twisted Bernoulli numbers (TBNs)

Definition

For any root of unity ξ , we define the *twisted Bernoulli numbers* by

$$\mathcal{H}(t; \xi) = \frac{1}{1 - \xi e^t} = \sum_{n=-1}^{\infty} \mathfrak{B}_n(\xi) \frac{t^n}{n!},$$

where we formally let $(-1)! = 1$.

Example

In the case $\xi = 1$, we have

$$\mathfrak{B}_{-1}(1) = -1, \quad \mathfrak{B}_n(1) = -\frac{B_{n+1}}{n+1} \quad (n \in \mathbb{N}_0).$$

In the case $\xi \neq 1$, we have

$$\mathfrak{B}_0(\xi) = \frac{1}{1 - \xi}, \quad \mathfrak{B}_1(\xi) = \frac{\xi}{(1 - \xi)^2}, \quad \mathfrak{B}_2(\xi) = \frac{\xi(\xi + 1)}{(1 - \xi)^3}, \dots$$

Twisted multiple Bernoulli numbers (TMBNs)

Definition

Let $r \in \mathbb{N}$, $\gamma_1, \dots, \gamma_r \in \mathbb{C}$, and $\xi_1, \dots, \xi_r \in \mathbb{C} \setminus \{1\}$ be roots of unity. Set

$$\mathcal{H}_r((t_j); (\xi_j); (\gamma_j)) := \prod_{j=1}^r \mathcal{H} \left(\gamma_j \sum_{k=j}^r t_k; \xi_j \right) = \prod_{j=1}^r \frac{1}{1 - \xi_j \exp(\gamma_j \sum_{k=j}^r t_k)}$$

and define *twisted multiple Bernoulli numbers* $\mathfrak{B}((n_j); (\xi_j); (\gamma_j))$ for $(n_j) \in \mathbb{Z}_{\geq 0}^r$ by

$$\mathcal{H}_r((t_j); (\xi_j); (\gamma_j)) = \sum_{n_1=0}^{\infty} \cdots \sum_{n_r=0}^{\infty} \mathfrak{B}((n_j); (\xi_j); (\gamma_j)) \frac{t_1^{n_1}}{n_1!} \cdots \frac{t_r^{n_r}}{n_r!}.$$

In the case $r = 1$, we have $\mathfrak{B}(n; \xi; 1) = \mathfrak{B}_n(\xi)$. In the general case, we use the notation $\mathfrak{B}_{(n_j)_r}((\xi_j)_r) := \mathfrak{B}((n_j)_r; (\xi_j)_r; (1)_r)$ when $\gamma_1 = \cdots = \gamma_r = 1$.

Explicit formula for CMBNs

Theorem (F., 2026+)

The following formula holds for every l, ξ .

$$\mathcal{B}_{l, \xi}^{((l_i)_r; (\epsilon_i)_r; (\kappa_i)_{r-1})} = \sum_{\mathbf{e}=(e_i)_r \in \{0\} \times \{0,1\}^{r-1}} \sum_{\substack{\forall 1 \leq i \leq j \leq r \\ m_{\mathbf{e}, i, j} \in \mathbb{N}_0 \\ m_{\mathbf{e}, 1, 1} \leq l_1 \\ \dots \\ m_{\mathbf{e}, 1, r} + \dots + m_{\mathbf{e}, r, r} \leq l_r}} \sum_{i_1=0}^{1-e_1} \cdots \sum_{i_r=0}^{1-e_r} \\ (-1)^{\sum_{k=1}^r (e_k + i_k)} \delta_{\xi, \prod_{k=1}^r \epsilon_k^{-(\kappa_{k-1} + i_k)}} \prod_{k=1}^r (\kappa_{k-1} + i_k)^{\sum_{j=k}^r m_{\mathbf{e}, k, j}} \\ \prod_{k=1}^r \binom{l_k}{m_{\mathbf{e}, 1, k}, \dots, m_{\mathbf{e}, k, k}, l_k - \left(\sum_{i=1}^k m_{\mathbf{e}, i, k}\right)} \delta_{l, \sum_{k=1}^r \sum_{j=k}^r m_{\mathbf{e}, k, j}} \\ \mathfrak{B}_{(l_1 - m_{\mathbf{e}, 1, 1} \circ_{e_2} \cdots \circ_{e_r} l_r - (\sum_{i=1}^r m_{\mathbf{e}, i, r}))} (\epsilon_1^{-1}(1 - e_1) \circ_{e_2} \cdots \circ_{e_r} \epsilon_r^{-1}(1 - e_r))$$

$$(\text{CMBNs}) = \sum (\text{TMBNs})$$

Main theorem (The case $r = 1$)

Example (The case $r = 1$)

For any root of unity ξ_1 , TBNs is defined by

$$\mathcal{H}(t; \xi_1) = \frac{1}{1 - \xi_1 e^t} = \sum_{n=-1}^{\infty} \mathfrak{B}_n(\xi_1) \frac{t^n}{n!}.$$

Let $c \in \mathbb{Z}_p^\times \cap \mathbb{N}_{\geq 2}$. For any $l_1 \in \mathbb{N}_0$, $\epsilon_1 \in \mu_c$, and $h \in \mathbb{N}$, CHSs is defined by

$$\mathcal{S}_{\emptyset, h}(l_1; \epsilon_1) = \sum_{u_1=0}^{h-1} \epsilon_1^{-u_1} u_1^{l_1}.$$

For any $\xi \in \mu_c$ and $l \in \{0, \dots, l_1 + 1\}$, CBNs is defined by

$$\mathcal{S}_{\emptyset, h}(l_1; \epsilon_1) = \sum_{\substack{l=0 \\ \xi \in \mu_c}}^{l_1+1} \mathcal{B}_{l, \xi}^{(l_1; \epsilon_1; \emptyset)} h^l \xi^h.$$

Main theorem (The case $r = 1$)

Theorem (The case $r = 1$)

Let $c \in \mathbb{Z}_p^\times \cap \mathbb{N}$, $c \geq 2$. Put $l_1 \in \mathbb{N}_0$ be a non-negative integer and $\epsilon_1 \in \mu_c$. Then, the following formula

$$\mathcal{B}_{l,\xi}^{(l_1;\epsilon_1;\emptyset)} = \mathfrak{B}_h(\epsilon_1^{-1})\delta_{\xi,1}\delta_{l,0} - \sum_{l'=0}^{l_1+1} \binom{l_1}{l'} \mathfrak{B}_{h-l'}(\epsilon_1^{-1})\delta_{\xi,\epsilon_1^{-1}}\delta_{l,l'}$$

holds for every ξ .

Proof.

Consider the following generating function with coefficient $\mathcal{S}_{\emptyset,h}(l_1;\epsilon_1)$

$$F(t) = \sum_{l_1=0}^{\infty} \mathcal{S}_{\emptyset,h}(l_1;\epsilon_1) \frac{t^{l_1}}{l_1!} = \sum_{l_1=0}^{\infty} \sum_{u_1=0}^{h-1} \epsilon_1^{-u_1} \frac{(u_1 t)^{l_1}}{l_1!}.$$

Main theorem (The case $r = 1$)

Proof (continued).

For convenience, we let $l_1 = l$, $u_1 = u$, and $\epsilon_1^{-1} = \xi$,

$$\begin{aligned} F(t) &= \sum_{l=0}^{\infty} \sum_{u=0}^{h-1} \xi^u \frac{(ut)^l}{l!} \\ &= \sum_{u=0}^{h-1} \xi^u \sum_{l=0}^{\infty} \frac{(ut)^l}{l!} \\ &= \sum_{u=0}^{h-1} \xi^u e^{ut} \\ &= \sum_{u=0}^{h-1} (\xi e^t)^u = \frac{1 - (\xi e^t)^h}{1 - \xi e^t}. \end{aligned}$$

Main theorem (The case $r = 1$)*Proof* (continued).

By definition of TBNs, we obtain

$$\begin{aligned}
 F(t) &= \frac{1 - (\xi e^t)^h}{1 - \xi e^t} = (1 - \xi^h e^{th}) \frac{1}{1 - \xi e^t} \\
 &= \left(1 - \xi^h \sum_{l=0}^{\infty} \frac{(th)^l}{l!} \right) \left(\sum_{n=-1}^{\infty} \mathfrak{B}_n(\xi) \frac{t^n}{n!} \right) \\
 &= \sum_{n=-1}^{\infty} \mathfrak{B}_n(\xi) \frac{t^n}{n!} - \left(\xi^h \sum_{l=0}^{\infty} h^l \frac{t^l}{l!} \right) \left(\sum_{n=-1}^{\infty} \mathfrak{B}_n(\xi) \frac{t^n}{n!} \right) \\
 &= \sum_{m=-1}^{\infty} \mathfrak{B}_m(\xi) \frac{t^m}{m!} - \xi^h \sum_{m=-1}^{\infty} \sum_{l+n=m} h^l \mathfrak{B}_n(\xi) \frac{t^m}{l!n!}.
 \end{aligned}$$

Main theorem (The case $r = 1$)*Proof (continued).*

$$\begin{aligned}
F(t) &= \sum_{m=-1}^{\infty} \mathfrak{B}_m(\xi) \frac{t^m}{m!} - \xi^h \sum_{m=-1}^{\infty} \sum_{l+n=m} h^l \mathfrak{B}_n(\xi) \frac{t^m}{l!n!} \\
&= \sum_{m=-1}^{\infty} \mathfrak{B}_m(\xi) \frac{t^m}{m!} - \xi^h \sum_{m=-1}^{\infty} \sum_{l=0}^{m+1} h^l \mathfrak{B}_{m-l}(\xi) \frac{t^m}{l!(m-l)!} \\
&= \sum_{m=-1}^{\infty} \mathfrak{B}_m(\xi) \frac{t^m}{m!} - \xi^h \sum_{m=-1}^{\infty} \sum_{l=0}^{m+1} h^l \mathfrak{B}_{m-l}(\xi) \binom{m}{l} \frac{t^m}{m!} \\
&= \sum_{m=-1}^{\infty} \left(\mathfrak{B}_m(\xi) - \xi^h \sum_{l=0}^{m+1} \binom{m}{l} h^l \mathfrak{B}_{m-l}(\xi) \right) \frac{t^m}{m!}.
\end{aligned}$$

Main theorem (The case $r = 1$)

Proof (continued).

By comparing the coefficients, we have

$$\mathcal{S}_{\emptyset, h}(l_1; \epsilon_1) = \mathfrak{B}_h(\epsilon_1^{-1}) - \sum_{l=0}^{l_1+1} \binom{l_1}{l} \mathfrak{B}_{h-l}(\epsilon_1^{-1}) \epsilon_1^{-h} h^l.$$

By comparing the coefficients, we obtain

$$\mathcal{B}_{l, \xi}^{(l_1; \epsilon_1; \emptyset)} = \mathfrak{B}_h(\epsilon_1^{-1}) \delta_{\xi, 1} \delta_{l, 0} - \sum_{l'=0}^{l_1+1} \binom{l_1}{l'} \mathfrak{B}_{h-l'}(\epsilon_1^{-1}) \delta_{\xi, \epsilon_1^{-1}} \delta_{l, l'},$$

where δ is the Kronecker delta. □

Application to p-adic special values

Generalized Euler-Zagier-Lerch type multiple zeta-function

Definition

Let $\xi_1, \dots, \xi_r \in \mathbb{C}$ be roots of unity. For $\gamma_1, \dots, \gamma_r \in \mathbb{C}$ with $\Re \gamma_j > 0$ ($1 \leq j \leq r$), where $\Re$ means the real part of the complex number. The *multiple zeta-function of the generalized Euler-Zagier-Lerch type* is defined by

$$\zeta_r((s_j); (\xi_j); (\gamma_j)) := \sum_{m_1=1}^{\infty} \cdots \sum_{m_r=1}^{\infty} \prod_{j=1}^r \frac{\xi_j^{m_j}}{(m_1 \gamma_1 + \cdots + m_j \gamma_j)^{s_j}},$$

which is absolutely convergent in the region

$$\mathcal{D}_r = \{(s_1, \dots, s_r) \in \mathbb{C}^r \mid \Re(s_{r-k+1} + \cdots + s_r) > k \ (1 \leq k \leq r)\}.$$

Furusho, Komori, Matsumoto, and Tsumura's work

Theorem (Furusho–Komori–Matsumoto–Tsumura, 2017)

Let $\xi_1, \dots, \xi_r \in \mathbb{C}$ be roots of unity and $\gamma_1, \dots, \gamma_r \in \mathbb{C}$ with $\Re \gamma_j > 0$ ($1 \leq j \leq r$). Assume that

$$\xi_j \neq 1 \text{ for all } j \ (1 \leq j \leq r).$$

Then, with the above notation, $\zeta_r((s_j); (\xi_j); (\gamma_j))$ can be analytically continued to $\mathbb{C}^r$ as an entire function in (s_j) . For $n_1, \dots, n_r \in \mathbb{N}_0$,

$$\zeta_r((-n_j); (\xi_j); (\gamma_j)) = (-1)^{r+n_1+\dots+n_r} \mathfrak{B}((n_j); (\xi_j^{-1}); (\gamma_j)).$$

Remark

In the case $\gamma_1 = \dots = \gamma_r = 1$, we have

$$\zeta_r((-n_j); (\xi_j); (1)) = (-1)^{r+n_1+\dots+n_r} \mathfrak{B}_{(n_j)_r}((\xi_j^{-1})_r).$$

An explicit formula for the p MLFVs

We have the following three results.

$$p\text{MLFVs} = \sum (\text{CMBNs}) \times (\text{CMHV}_s) \text{ [FJ]}$$

$$(\text{CMBNs}) = \sum (\text{TMBNs}) \text{ (F.)}$$

$$\text{zeta functions of } (-n_j) = \text{TMBNs of } (n_j) \text{ [FKMT]}$$

By the above, we obtain

$$\begin{aligned} p\text{MLFVs} &= \sum (\text{CMBNs}) \times (\text{CMHV}_s) \text{ [FJ]} \\ &= \sum \left(\sum \text{TMBNs} \right) \times (\text{CMHV}_s) \text{ (F.)} \\ &= \sum (\text{complex zeta values}) \times (\text{CMHV}_s) \text{ [FKMT]} \end{aligned}$$

An explicit formula for the p MLFVs

Theorem (F., 2026+)

$$\begin{aligned}
 & \left(p^{\sum_{i=1}^r n_i} L_{p,r}((n_i)_r; (\omega^{-n_i})_r; (1)_r; c) \right)_{p \in \mathcal{P}_c} = \\
 & \sum_{l=(l_i)_r \in \mathbb{N}_0^r} \sum_{\substack{\epsilon \in \mu_c \\ (\epsilon_i)_r \in (\mu_c \setminus \{1\})^r}} \sum_{J=(P_1, P_2, P_3) \in E_r} \frac{\epsilon}{\prod_{i=1}^r (1 - \epsilon_i)} \prod_{i=1}^r \binom{-n_i}{l_i} \\
 & \sum_{(e_i)_{\sharp(P_2)} \in \{0\} \times \{0,1\}^{\sharp(P_2)-1}} \sum_{i_1=0}^{1-e_1} \cdots \sum_{i_{\sharp(P_2)=0}}^{1-e_{\sharp(P_2)}} (-1)^{\sharp(P_2) + \sum_{k=1}^{\sharp(P_2)} (i_k + l_{J,k})} \\
 & \zeta_{\sharp(P_2) - \sum_{k=1}^{\sharp(P_2)} e_k} \left((-l_{J,1} \circ_{e_2} \cdots \circ_{e_{\sharp(P_2)}} -l_{J,\sharp(P_2)}); \right. \\
 & \left. (\epsilon_{J,1}(1 - e_1) \circ_{e_2} \cdots \circ_{e_{\sharp(P_2)}} \epsilon_{J,\sharp(P_2)}(1 - e_{\sharp(P_2)})) \right) \\
 & \delta \sum_{\epsilon \in \prod_{j \in P_3} \left(\frac{\epsilon_j^{(P_2,3)}}{\epsilon_j} \right)^{\kappa_j'}} \prod_{k=1}^{\sharp(P_2)} \epsilon_{J,k}^{-(\kappa_{J,k-1} + i_k)} \sum_{\delta \in \Delta(T_{r,J})} \mathfrak{H}(w(l)_\delta)^{\text{Frob}^{-1}}.
 \end{aligned}$$

Depth 1 formula for the p MLFVs

Corollary (F., 2026+)

For any $n \in \mathbb{N}$, the family

$$(p^n L_{p,1}(n; \omega^{-n}; 1; c))_{p \in \mathcal{P}_c}$$

is expressed as

$$\sum_{l=0}^{\infty} \sum_{\epsilon \in \mu_c \setminus \{1\}} \binom{-n}{l} (-1)^{1+l} \zeta_1(-l; \epsilon^{-1}; 1) \mathfrak{H}(n+l, \epsilon^{-1})^{\text{Frob}^{-1}}.$$

Depth 2 formula for the p MLFVs

Corollary (F., 2026+)

For any $(n_1, n_2) \in \mathbb{N}^2$, the family

$$(p^{n_1+n_2} L_{p,2}((n_1, n_2); (\omega^{-n_1}, \omega^{-n_2}); (1, 1); c))_{p \in \mathcal{P}_c}$$

can be expressed as

$$\begin{aligned} & \sum_{(l_1, l_2) \in \mathbb{N}_0^2} \sum_{(\epsilon_1, \epsilon_2) \in (\mu_c \setminus \{1\})^2} \binom{-n_1}{l_1} \binom{-n_2}{l_2} (-1)^{l_1+l_2} \\ & \left\{ \zeta_2((-l_1, -l_2); (\epsilon_1^{-1}, \epsilon_2^{-1})) \mathfrak{H}((n_1 + l_1, n_2 + l_2), (\epsilon_1^{-1}, \epsilon_2^{-1}))^{\text{Frob}^{-1}} \right. \\ & \quad + \zeta_2((-l_1, -l_2); (\epsilon_1^{-1}, \epsilon_2^{-1})) \mathfrak{H}(n_1 + l_1 + n_2 + l_2, \epsilon_1^{-1})^{\text{Frob}^{-1}} \\ & \quad + \zeta_2((-l_1, -l_2); (\epsilon_1^{-1}, \epsilon_2^{-1})) \mathfrak{H}((n_2 + l_2, n_1 + l_1), (\epsilon_1^{-1}, \epsilon_1^{-1} \epsilon_2)) ^{\text{Frob}^{-1}} \\ & \quad \left. + \zeta_1(-l_1 - l_2; \epsilon_1^{-1}) \mathfrak{H}((n_2 + l_2, n_1 + l_1), (\epsilon_1^{-1}, \epsilon_1^{-1} \epsilon_2)) ^{\text{Frob}^{-1}} \right\}. \end{aligned}$$

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Thank you for your attention!

Appendix

p -adic multiple L -functions

Definition

Fix a prime number p , we put $\mathcal{O}_{\mathbb{C}_p}$ to be the ring of integers of $\mathbb{C}_p$. Let $\omega : \mathcal{O}_{\mathbb{C}_p}^\times \rightarrow \mathcal{O}_{\mathbb{C}_p}^\times$ be the Teichmüller character, and let $\langle x \rangle = \frac{x}{\omega(x)}$ for $x \in \mathcal{O}_{\mathbb{C}_p}^\times$. Set

$$\int_{\mathbb{Z}_p} f(x) d\mathbf{m}_z(x) = \lim_{N \rightarrow \infty} \sum_{a=0}^{p^N-1} f(a) \mathbf{m}_z(a + p^N \mathbb{Z}_p),$$

where $\mathbf{m}_z$ is the measure defined by

$$\mathbf{m}_z(j + p^N \mathbb{Z}_p) = \frac{z^j}{1 - z^{p^N}} \quad (0 \leq j \leq p^N - 1)$$

for $z \in \mathbb{P}^1(\mathbb{C}_p)$ with $|z - 1|_p \geq 1$.

p -adic multiple L -functions

Definition (continued)

Let p be a prime number and $(s_i)_r$ be the element in the set

$$\mathfrak{X}_r(d) := \{(s_1, \dots, s_r) \in \mathbb{C}_p^r \mid |s_j|_p \leq d^{-1} p^{-\frac{1}{p-1}} \ (1 \leq j \leq r)\}.$$

Let $(k_i)_r \in \mathbb{Z}^r$, $c \in \mathbb{N}_{>1}$ which is prime to p , and

$$(\mathbb{Z}_p^r)' := \{(x_1, \dots, x_r) \in \mathbb{Z}_p^r \mid p \nmid x_1, \ p \nmid (x_1 + x_2), \dots, \ p \nmid (x_1 + \dots + x_r)\}.$$

p -adic multiple L -functions

Definition (continued)

The p -adic multiple L -function is defined by

$$L_{p,r}((s_i)_r; (\omega^{k_i})_r; (1)_r; c) := \int_{(\mathbb{Z}_p^r)'} \langle x_1 \rangle^{-s_1} \langle x_1 + x_2 \rangle^{-s_2} \cdots \langle x_1 + \cdots + x_r \rangle^{-s_r} \omega^{k_1}(x_1) \omega^{k_2}(x_1 + x_2) \cdots \omega^{k_r}(x_1 + \cdots + x_r) \prod_{i=1}^r d\tilde{m}_c(x_i),$$

where $\tilde{m}_c$ is the measure defined by

$$\tilde{m}_c := \sum_{\substack{\xi^c=1 \\ \xi \neq 1}} m_\xi.$$